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Bubble size distributions in a bubble column reactor under industrial conditions

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Abstract

The design of gas liquid processes requires a detailed knowledge of bubble size distributions, since they determine the mass transfer. In this contribution the influence of operating conditions and physical properties of gas and liquid phase on initial and final ("stable") bubble sizes is discussed. The measurements were performed in a lab scale bubble column for different liquids sparged with nitrogen for pressures up to 50 bars and temperatures up to 175 °C. Bubble size distributions were determined by image processing.

Bubbles tend to become smaller with decreasing surface tension, increasing gas density and decreasing liquid viscosity, resulting in reduced stable bubble sizes at increased pressure and also at increased temperature as long as evaporation can be neglected. Impurities in aqueous and organic liquids can severely influence bubble sizes by restraining coalescence. For such systems bubble size distributions in a column mainly depend on the size of the primary bubbles, which are determined by the sparger design.
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Keywords: Bubble breakage; Bubble column; Bubble formation; Coalescence; High pressure and temperature; Stable Sauter-diameter

1. Introduction

Most investigations in bubble columns have been performed at ambient conditions and for aqueous systems in spite of the fact that industrial bubble columns are often operated at increased temperature and pressure using organic liquids.

Recent findings on bubble size distributions at fixed positions in the reactors show that bubble sizes are influenced significantly by pressure and temperature [13,23]. Investigations on primary bubble sizes show, that for aqueous systems the initial bubble size decreases with increased pressure [7,9,10,21,24]. Luo et al. [14] measured bubble size distributions in a bubble column at ≈0.3 m above the sparger for nitrogen bubbles in Paratherm NF heat transfer fluid and also found decreasing bubble sizes with increasing pressure. Similar results have been obtained by Lin et al. [13], who determined bubble size distributions for the same gas liquid system 0.4 m above the sparger. They further measured physical properties in situ and found decreasing bubble sizes with rising temperature. In contrast to Lin et al. [13], Pohorecki et al. [17] found no dependence on temperature and pressure. They published bubble size distributions of nitrogen bubbles in water for different height levels of a bubble column determined from photographs for varying pressure, temperature, superficial gas velocity and for different spargers, stating that the average bubble diameter is practically constant to a value of 6.67 mm (±12%). Recently Pohorecki et al. found considerably smaller bubble sizes for N_2/cyclohexane [18], which they correlated with superficial gas velocity by $d_S = 0.001658 u_G^{\sup\ -0.12}$.

These contradictory examinations probably result from differences in the column design, sparger configurations or measuring techniques. In order to quantify the effect of physical properties such as gas density, liquid viscosity, surface tension and of operating conditions such as pressure, temperature, superficial gas velocity and of liquid impurities on bubble size distributions, detailed measurements have been performed.

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Nomenclature

A_i^c	cross-sectional area of bubble (m^2)
d_i	spheric bubble diameter (m)
d_S	Sauter mean bubble diameter (m)
d_S^{ini}	initial Sauter mean bubble diameter (m)
d_S^{∞}	stable Sauter mean bubble diameter (m)
d_{sp}	sparging diameter (pore size, inner diameter of ring or capillary sparger) (m)
E_σ	surface energy (J)
f_N	distribution density function (1/mm)
p	system pressure (bar)
T	temperature (°C)
u_G^{sup}	superficial gas velocity (cm/s)
V_{S1}	volume of spherical bubble corresponding to Sauter mean bubble diameter at lower column section (m^3)
V_{S5}	volume of spherical bubble corresponding to Sauter mean bubble diameter at upper column section (m^3)
Greeks	
ε_G	gas holdup (–)
γ	contact angle (°)
η_L	liquid viscosity (kg/m s)
ρ_G	gas density (kg/m^3)
σ	surface tension (N/m)

2. Experimental

Bubble size distributions were determined by non-invasive image processing. Due to the transparent bubble column design with windows from 4 sides (Fig. 1, left) measurements across the complete cross section at different levels could be performed. Bubbly flow in a glass tube inside the reactor casing with an inner diameter of 54 mm was investigated at pressures up to 50 bars and temperatures up to 175 °C for varying superficial gas velocities and different N_2/liquid systems. Only one parameter was varied at a time. Measurements were performed in the homogeneous regime, granting constant flow conditions. In order to ensure well defined sparging, most experiments were performed with a special capillary sparger, which consists of 19 capillaries of 25 mm in length with an inner diameter of 0.15 mm, granting well defined pressure drop for each sparging orifice. Thus gas could be sparged uniformly into the liquid phase generating narrow primary bubble size distributions. Fig. 1 shows the experimental set-up.

In order to determine the bubble size distribution at a specific column level, images of the bubbly flow were taken by a CCD-camera. For getting a sufficiently representative bubble size distribution usually sizes of at least 1000 bubbles were determined. Good image quality was obtained when illuminating the bubbles with diffuse back light (Fig. 2). Bubbles then appear dark on white background with a bright spot in the middle (Fig. 2). This spot was used in the image processing step to

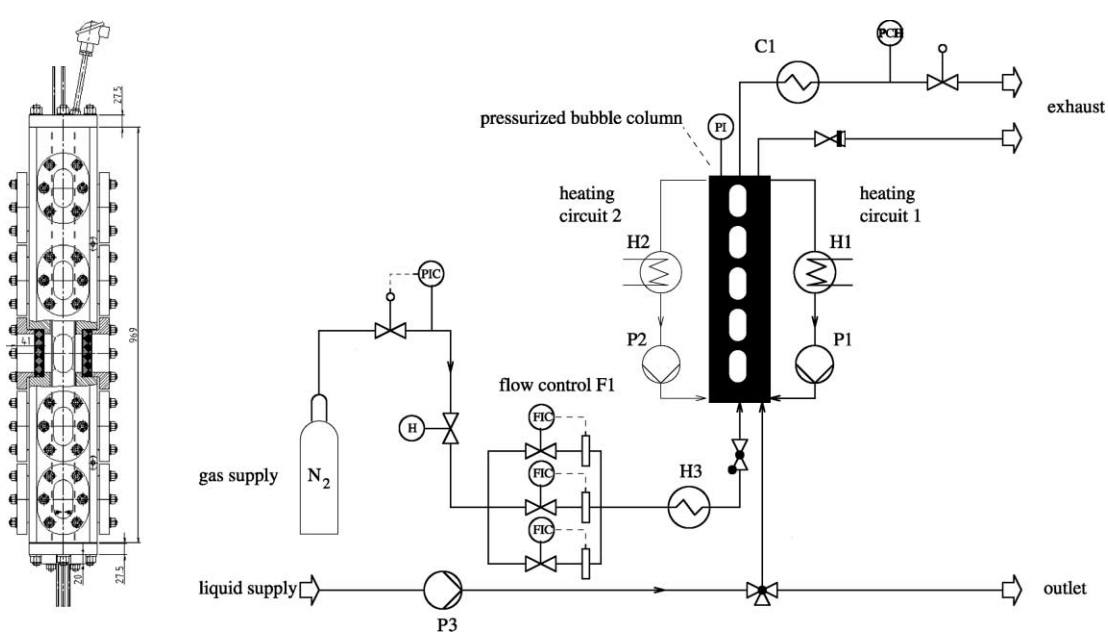


Fig. 1. Left: pressurized bubble column. Right: scheme of the experimental set-up.

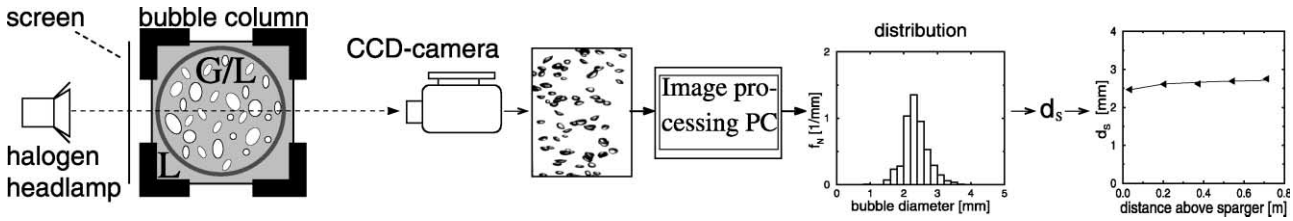


Fig. 2. Scheme for determination of bubble size distributions by image processing.

eliminate overlapping bubbles in order to avoid their interpretation as one big bubble. Otherwise the method of image processing corresponds to the approach developed by Borchers and Eigenberger [1]. For the estimation of bubble sizes the cross-sectional area A_i^c of all selected non-overlapping bubbles is determined. The diameters of corresponding spherical bubbles $d_i = \sqrt{4A_i^c/\pi}$ are calculated from the cross-sectional area. Under the conditions discussed in the present work, bubble sizes were usually less than 3.5 mm in spheric diameter. Bubble size measurements can be reproduced within ± 0.05 mm (Fig. 13).

A certain inaccuracy has to be attributed to the fact that bubbles are not spherical but deformed, especially for larger bubble sizes. Bubbles flatten due to the drag forces and can be approximated by a disk like rotational symmetric ellipsoid [5]. For a horizontal disk, the calculation of the corresponding spheric diameter from the projection area leads to an underestimation of the bubble size. Usually disk shaped bubbles are inclined with an average inclination angle of 25° [15], which renders the projection area larger when inclined towards the camera, compensating part of the underprediction. If the larger elliptic axis is twice as long as the smaller elliptic axis, the relative error lies in between -1% and -11% depending on the perspective [2], and leads to an average underprediction of the size of disk shaped bubbles by about 6%. Smaller bubbles are less flattened, hence the deviation is reduced.

For mass-transfer calculations the interfacial area is an important factor. For approximate calculations the Sauter diameter $d_S = \sum_{i=1}^{N} d_i^3 / \sum_{i=1}^{N} d_i^2$ can be used. In order to study the development of the bubble size distributions with increasing distance from the sparger, the Sauter diameter has been determined for different levels of the bubble column. At sufficient height the Sauter diameter should approach the stable bubble diameter d_S^∞, which results from the equilibrium of bubble breakage and coalescence.

3. Results and discussion

3.1. Stable bubble diameter

Influences of physical properties on stable bubble sizes were determined by comparing bubble sizes of similar gas liquid systems, which differ in one physical property.

3.1.1. Influence of physical properties on bubble size distribution

3.1.1.1. Influence of gas density. The influence of gas density and operating pressure on stable bubble sizes is closely connected, since pressure mainly affects gas density. Idogawa et al. and Wilkinson [6,22] investigated gas/water systems for different gases and variable pressure and found decreased bubble sizes for increasing gas density. Wilkinson explained this behavior with increased bubble breakage. Other investigators found the same, but at increased pressure this effect was found to be less pronounced [4]. Coalescence investigations of Sagert and Quinn [19,20] confirmed this trend. For increased pressure they observed increased coalescence times for CO_2, H_2S, N_2O and alkane bubble pairs in water, hence coalescence is declined.

To study the influence of pressure on bubble size distributions, several experiments with nitrogen bubbles in water were performed.

- In a stagnant coalescence experiment two nitrogen bubbles emerging slowly from two opposing nozzles were brought into contact at 1 and 45 bar. The coalescence time increased only slightly with pressure.
- In a dynamic coalescence experiment a rising bubble could collide with a second one, attached to a nozzle. In this experiment the coalescence probability decreased significantly with increasing pressure.
- A sequence of single bubbles was detached from a single nozzle and its subsequent breakage was recorded (Fig. 3). This experiment showed an increase of breakage probability with increasing pressure.
- Bubble swarms were generated using different spargers and the development of the mean Sauter diameter was recorded with increasing distance from the sparger (Fig. 4). These experiments showed an increase of the Sauter diameter towards a stable bubble diameter which is decreased with increasing pressure.

This leads to the following interpretation: Pressure increase leads to an increase in bubble breakage due to the increase in gas density, caused by the larger inertia of the gas in the fluctuating bubble. This is clearly shown

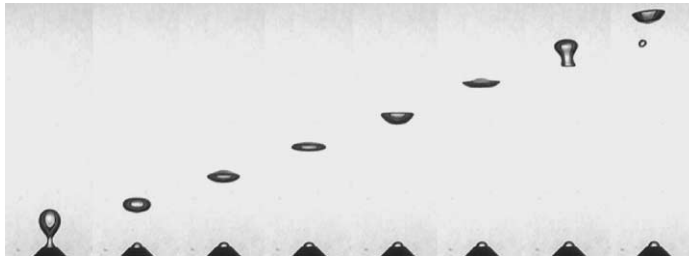


Fig. 3. Bubble breakage probabilities of single bubbles detaching from single nozzle sparger with $d_{sp} = 2$ mm for ambient and high pressure ($T = 25$ °C).

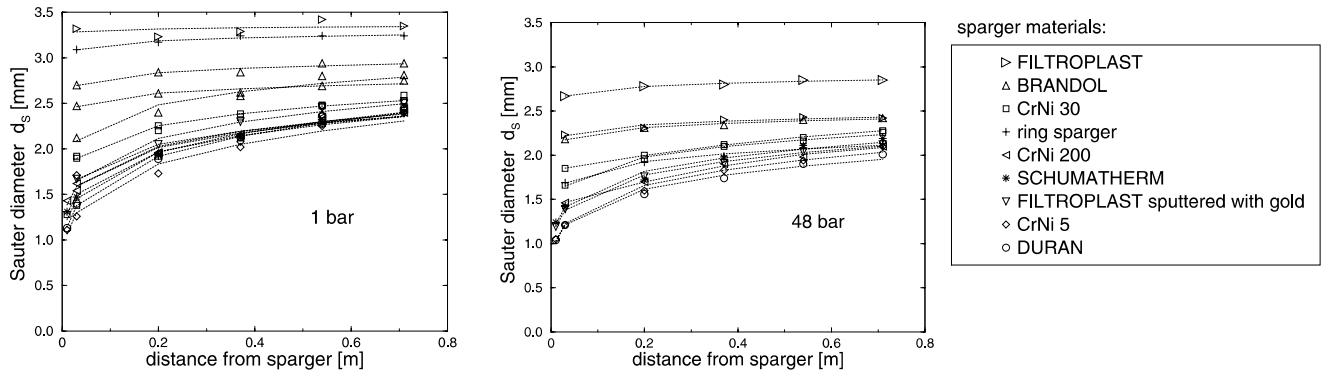


Fig. 4. Comparison of Sauter diameters as a function of the distance from sparger for different initial bubble diameters at 1 and 48 bar (N_2/water, $u_G^{sup} \approx 0.25$ cm/s, 20 °C, porous and ring sparger).

by the experiment of single bubble detachments. Surface tension decreases in the above experiments between 1 and 45 bar by 5.4% and hence has an almost negligible influence on bubble coalescence. This could be seen in the stagnant coalescence experiments.

In bubble swarms the combined influence of coalescence and breakage leads to the observed increase of bubble size over column height (Fig. 4). Since coalescence is almost unaffected by pressure but breakage is increased, the mean bubble size decreases with increasing pressure. From the experiments with different spargers it can be concluded that in reality a band width of stable bubble sizes is created rather than a single stable bubble size which could be determined from breakage experiments [4].

3.1.1.2. Influence of surface tension. Surface energy increases with decreasing bubble size. Hence bubbles tend to coalesce as long as the gain in surface energy is not compensated by increased drag and inertial forces causing bubble breakage. For liquids with comparable physical properties lower surface tension results in smaller bubble sizes. This is shown by the comparison of the N_2/water system with N_2/cyclohexane at ambient conditions (Fig. 5, left). For both systems viscosity is nearly the same and liquid densities are comparable, but surface tension of cyclohexane is almost only one third of water (Table 1). Surface tension of ethanol is even lower and so is the stable Sauter diameter (Fig. 5, left). At increased pressure the stable bubble size is reduced (Fig. 5, left).

3.1.1.3. Influence of liquid viscosity. Rising liquid viscosity reduces turbulence in the liquid phase. Hence the energy of eddies is reduced and bubble breakage damped, leading to increased bubble sizes. This is illustrated by the comparison of bubble sizes of nitrogen bubbles in cyclohexanol, cyclohexanone and cyclohexane (Fig. 5, right), which differ at ambient condition mainly with respect to viscosity (Table 2). In highly viscous cyclohexanol bubbles are bigger than in cyclohexanone and cyclohexane. A similar behavior is seen at increased pressure, where bubble sizes are generally decreased.

3.1.2. Influence of operating conditions

The impact of changing temperature on stable bubble sizes is given by the temperature influence of the respective physical properties. An increase in temperature reduces both liquid viscosity and surface tension significantly, while its influence on gas density is less pronounced. Therefore increased temperature tends to

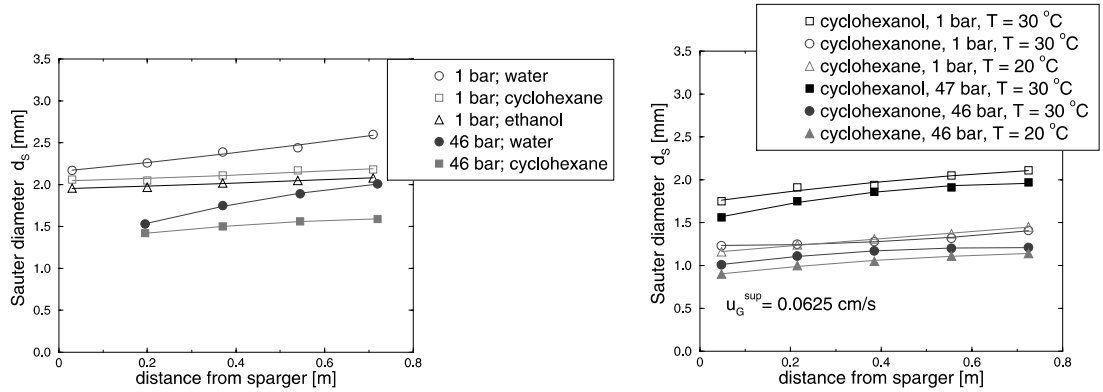


Fig. 5. Comparison of bubble sizes for liquids with differing surface tension for $u_G^{sup} = 0.25$ cm/s, $T = 20$ °C (left) and for liquids with different liquid viscosity for $u_G^{sup} = 0.0625$ cm/s (right).

Table 1
Properties of water, cyclohexane and ethanol at ambient conditions

Property (20 °C, 1 bar)	Water	Cyclohexane	Ethanol
ρ_L (kg/m^3)	998.2	778.5	789.3
η_L (10^{-3} kg/m s)	1.00	0.98	1.19
σ (mN/m)	72.75	24.98	22.03

Table 2
Properties of cyclohexane, cyclohexanone and cyclohexanol at ambient conditions

Property (1 bar)	Cyclohexane	Cyclohexanone	Cyclohexanol
ρ_L (kg/m^3)	778.5 (20 °C)	945.5 (20 °C)	949.3 (20 °C)
η_L (10^{-3} kg/m s)	0.98 (20 °C)	1.80 (30 °C)	41.07 (30 °C)
σ (mN/m)	24.98 (20 °C)	33.51 (30 °C)	33.47 (30 °C)

decrease stable bubble sizes, as long as operating pressure is high compared to saturation pressure and evaporation negligible (Fig. 6). However, for saturation pressures close to operation pressure, evaporation of liquid into the initially dry gas becomes more important and increases bubble sizes until saturation is reached. High pressure reduces bubble sizes due to increased gas density. Figs. 5 and 6 (right) show that this not only applies to aqueous systems but to hydrocarbon systems as well, as long as liquid viscosity changes little.

For Paratherm NF heat transfer fluid Lin et al. [13] found a similar influence of temperature and pressure on bubble sizes 0.3 m above the sparger. For ambient temperature as opposed to 78 °C the pressure influence was reduced, which can be explained by increased liquid viscosity.

Gas flow rate influences bubble number density and hydrodynamics. For the considered lab scale bubble column, LDA-measurements showed variations of liquid flow to be small, which can be explained by the limited size of the column. Increased superficial gas velocity increases the bubble collision frequency leading to a higher coalescence rate and an increased stable bubble diameter (Fig. 7). It also influences the rate at which bubbles approach the stable bubble size.

3.1.3. Influence of liquid impurities

Impurities causing coalescence inhibition can change bubble size distributions in column reactors drastically

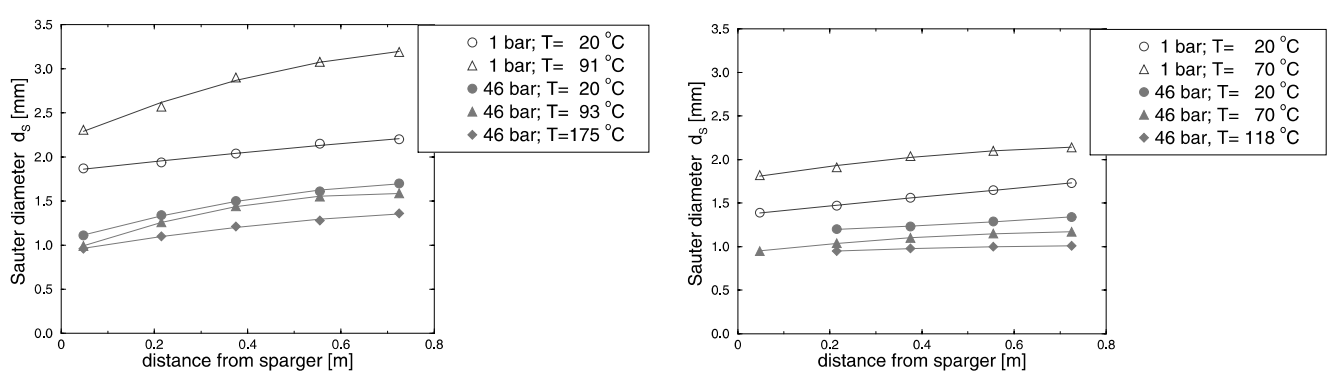
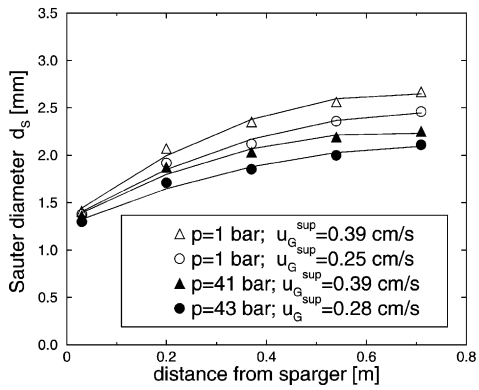


Fig. 6. Temperature and pressure influence on bubble sizes for N_2/water with $u_G^{sup} = 0.125$ cm/s (left) and for N_2/cyclohexane with $u_G^{sup} = 0.125$ cm/s (right).



V_{S5}/V_{S1}	1 bar	48 bar
u_G^{sup} = 0.25 cm /s	5.66	4.28
u_G^{sup} = 0.39 cm /s	6.79	4.53

Fig. 7. Left: influence of superficial gas velocity and pressure on bubble sizes (N_2/H_2O, $T = 20$ °C, porous CrNi sparger $d_{sp} = 32$ μm). Right: relative increase of bubble volume at 71 cm distance from the sparger compared to initial bubble volume V_{S5}/V_{S1}.

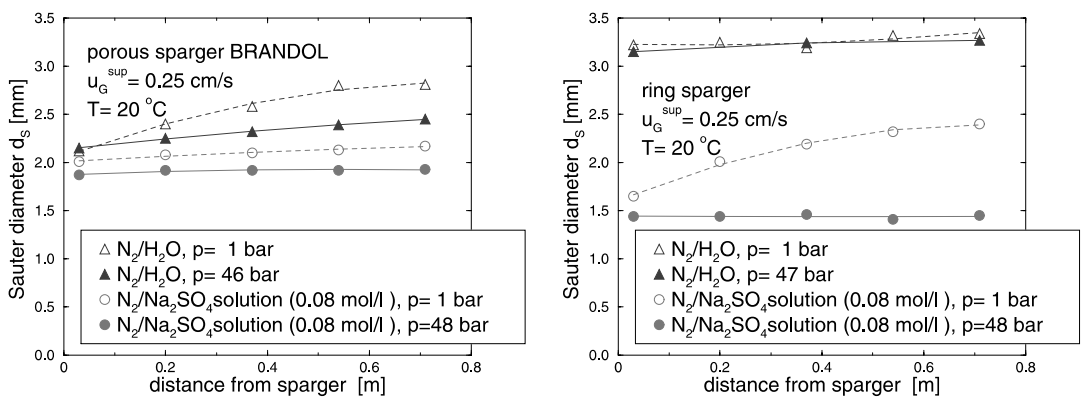


Fig. 8. Comparison of Sauter diameters as a function of the distance from sparger under coalescing and non-coalescing conditions for porous (left) and ring sparger (right).

(Fig. 8). This is already well known for aqueous systems [3,8,11,16]. Coalescence in aqueous systems is increasingly inhibited with rising concentrations of liquid impurities such as ions and hydrocarbons. For hydrocarbons coalescence inhibition is explained by the formation of a boundary layer at the gas/liquid interface and increases with rising polarity [3,8]. The polar group of the molecule interacts with the water while the hydrophobic group of the molecule is directed to the gas phase.

However, little is known for coalescence inhibition in organic reaction mixtures. As an example the system N_2/cyclohexane was studied. Compared to bubble sizes in pure cyclohexane for given superficial gas velocity the increase of the mean Sauter diameter with rising distance above the sparger is retarded when polar liquids such as cyclohexanol or cyclohexanone are added to cyclohexane (Fig. 9). This is especially the case for cyclohexanol, which is more polar than cyclohexanone. Coalescence inhibition can be explained similarily as for water, while the orientation of the molecules at the gas liquid interface has to be inverted.

3.2. Primary bubble size distributions

As shown in the last section, the sparger configuration is essential for non-coalescing systems as it determines bubble sizes in the whole bubble column (Figs. 8 and 9). For coalescing gas liquid systems the influence of the sparger configuration on bubble size distributions in large columns is restricted mainly to a smaller area above the spargers. Primary bubble size distributions depend on sparger configuration, operating conditions as well as on gas and liquid phase properties. They will be discussed for porous spargers, and ring and capillary sparger separately.

3.2.1. Porous sparger

Investigations of porous spargers have been performed for N_2/H_2O and N_2/aqueous Na_2SO_4 solution,

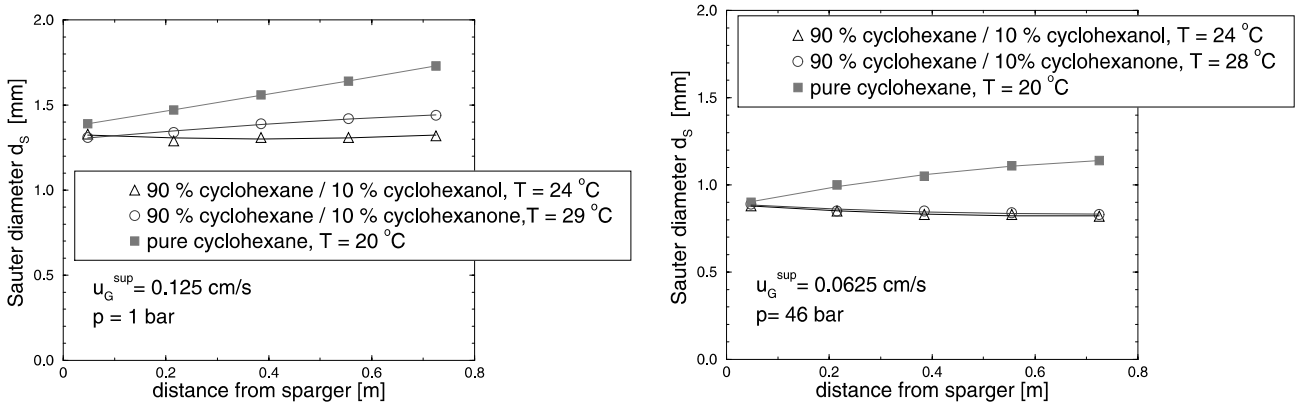


Fig. 9. Coalescence inhibition for a N_2/cyclohexane system due to addition of cyclohexanol and cyclohexanone at 1 bar for $u_G^{sup} = 0.125$ cm/s (left) and at 46 bar for $u_G^{sup} = 0.0625$ cm/s (right).

which is a coalescence inhibited system. Porous spargers showed little influence of operation conditions on initial bubble sizes. For gas velocities ranging from 0.25 to 0.4 cm/s no influence of the superficial gas velocity on the initial bubble diameter was observed (Fig. 11, left). Pressure tended to decrease initial bubble sizes slightly (Fig. 11, left).

The influence of pore size on initial bubble size distributions was studied for different chrome nickel spargers with average pore sizes of 70, 32 and 11 μm. No significant influence of pore sizes on initial bubble diameters was seen for the coalescing system N_2/H_2O. This is explained with retarded detachment of tiny bubbles, which coalesce on the surface of the porous plate until a certain bubble size is reached—provided that turbulence on the sparger surface is limited. For non-coalescing systems the initial bubble sizes correspond to the pore diameters, hence small bubbles with good mass transfer can be generated by fine porous spargers.

For coalescing systems the material of the sparger has the main influence on primary bubble sizes. For materials with good wettability (i.e. small contact angle), smaller initial bubbles are generated as compared with materials with big contact angle. This shows the comparison of initial bubble sizes generated by a polyethylene sparger with those generated by an identical polyethylene sparger sputtered with a gauzy layer of gold (Fig. 10). Similar results were obtained by Lin et al. [12]. As surface of materials can change due to calcification, precipitate, rust or other corrosion, initial bubble sizes can vary with time and therefore the reproducibility of experiments with porous spargers is bad.

3.2.2. Ring and capillary sparger

Better reproducibilities are reached with ring spargers or capillary spargers, which show similar characteristics. Initial bubble sizes of ring spargers and capillary spargers are influenced significantly by pressure (Figs. 8, right and 11, right). Increased pressure increases the momentum of the gas jets, which enforces turbulence and decreases initial bubble sizes. Coalescence inhibition showed little influence on primary bubble size distributions (Fig. 8).

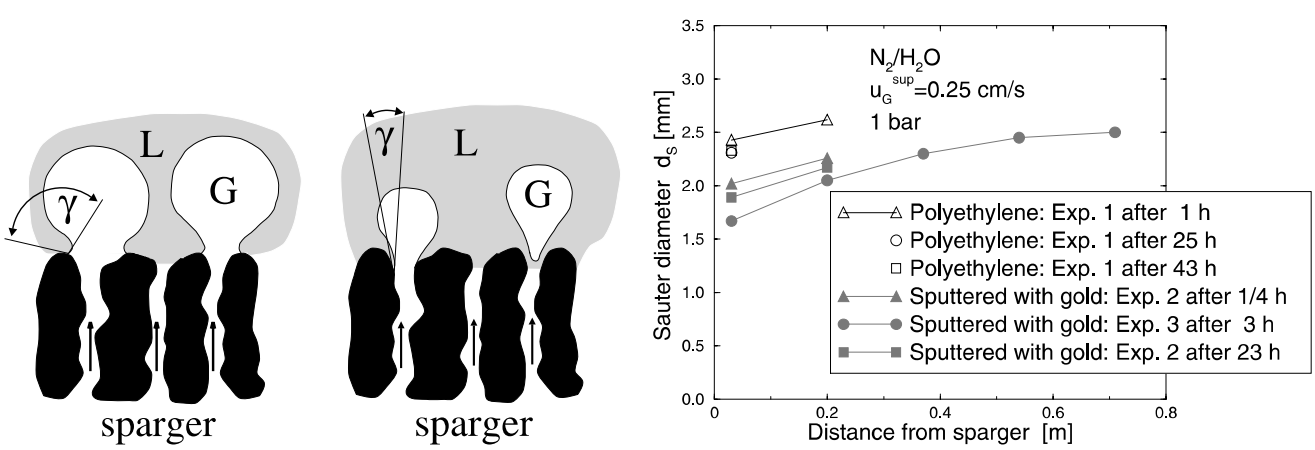


Fig. 10. Left/center: influence of contact angle on initial bubble diameter. Right: comparison of initial bubble diameters for polyethylene sparger without (left) and with (center) sputtered gold layer at atmospheric pressure and for $u_G^{sup} = 0.25$ cm/s ($T = 20$ °C).

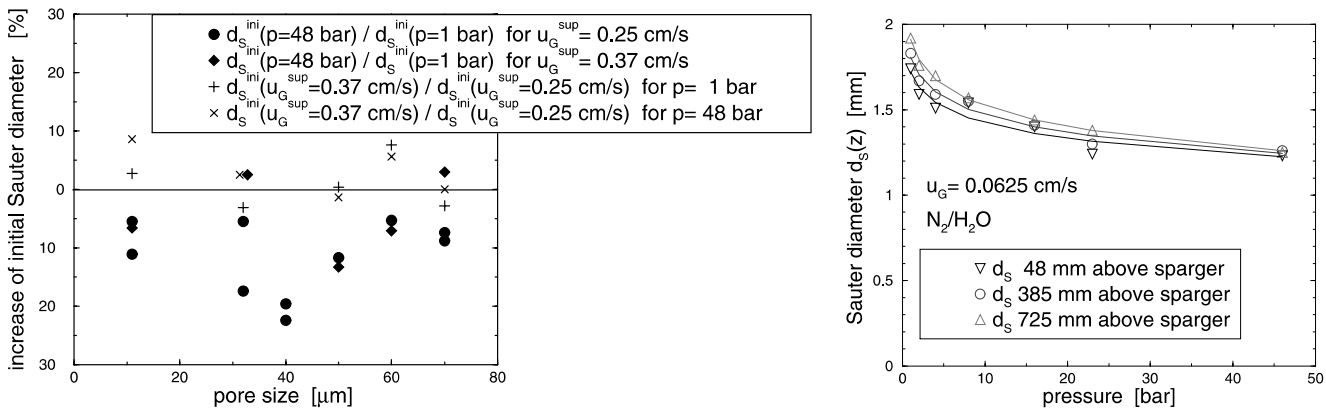


Fig. 11. Left: influence of pressure (●, ◆) and superficial gas velocity (+, ×) on initial bubble diameter in single experiments with different porous spargers (N_2/H_2O, $T = 20$ °C). Right: primary bubble sizes for capillary sparger with increasing pressure at low superficial gas velocity (capillary sparger with inner diameter $d_{sp} = 0.15$ mm).

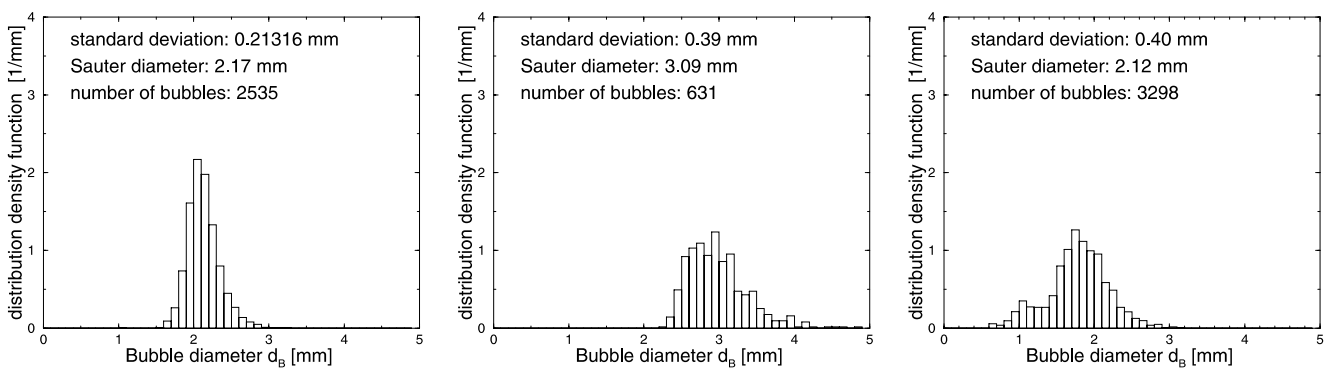


Fig. 12. Comparison of initial bubble size distributions of capillary sparger ($d_{sp} = 150$ μm), ring sparger ($d_{sp} = 400$ μm) and porous BRANDOL sparger ($d_{sp} = 50$ μm) with N_2/water at 1 bar and T= 20 °C for $u_G^{sup} \approx 0.25$ cm/s.

Capillary spargers generate bubbles even more reproducible than ring spargers, since the pressure drop of each sparging hole is well defined and almost the same. Compared to ring and porous spargers they generate very narrow initial bubble size distributions, hence the standard deviation is low (Fig. 12) and good reproducibility is obtained (Fig. 13). Bubble sizes are related to the inner diameter of the capillary. With tiny inner diameters extremely small bubbles less than 1 mm in diameter can be generated.

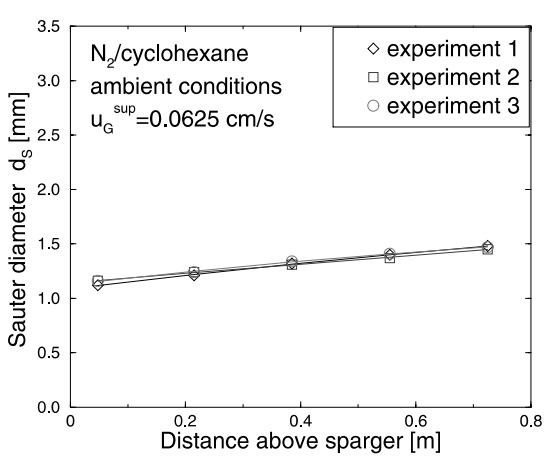


Fig. 13. Reproducibility of bubble sizes for a N_2/cyclohexane system sparged with capillary sparger for fixed operating conditions.

4. Conclusions

Bubble size distributions in a bubble column are determined mainly by three factors. The *primary bubble size distribution* determines the bubble size in the sparger region and depends on sparger configuration and material, liquid phase and operating conditions. Depending on the inclination of the system towards *coalescence and breakage*, bubbles reach the *stable bubble size distribution* after different rising times, which mainly depend on operating conditions and liquid phase properties. For large initial bubble diameters breakage is dominant and for small diameters coalescence. A hysteresis depending on the inclination of the system to coalescence or

Table 3
Influence of physical properties on stable Sauter diameter

$\rho_G \uparrow$	Inertia forces $\uparrow$	$\Rightarrow d_S^\infty \downarrow$
$\eta_L \downarrow$	Turbulence $\uparrow$	$\Rightarrow d_S^\infty \downarrow$
$\sigma \downarrow$	$E_\sigma \downarrow$	$\Rightarrow d_S^\infty \downarrow$

Table 4
Influence of operating conditions on stable Sauter diameter

p $\uparrow$	$\rho_G \uparrow$	$\eta_L -$	$\sigma \searrow$	$d_S^\infty \downarrow$
T $\uparrow$	$\rho_G \searrow$	$\eta_L \downarrow$	$\sigma \downarrow$	$d_S^\infty \downarrow$
	Evaporation $\uparrow$			$d_S^\infty \uparrow$
$u_G^{sup} \uparrow$	Bubble density $\uparrow$			$d_S^\infty \uparrow$

Table 5
Influence of pressure, pore size or orifice diameter, sparger material and grade of coalescence inhibition on initial Sauter diameter at $T = 20$ °C for ring, capillary and porous sparger

Changed parameter	$p \uparrow$	$d_{sp} \downarrow$	Contact angle $\uparrow$	Coalescence inclination $\downarrow$
Ring and capillary sparger	$d_S^{ini} \downarrow$	$d_S^{ini} \downarrow$	Not investigated	No influence on d_S^{ini}
Porous sparger	$d_S^{ini} \searrow$	$d_S^{ini} -$	$d_S^{ini} \uparrow$	$d_S^{ini} \downarrow$ for small pores

breakage between an upper and lower stable bubble diameter generally exists.

The stable bubble size is mainly influenced by gas density, liquid viscosity and surface tension (Table 3). Increased gas density increases inertial forces in the bubbles, favoring bubble breakage. Rising liquid viscosity reduces turbulence in the liquid phase. Hence the energy of eddies is reduced as well as bubble collision frequencies, hindering both bubble breakage and coalescence. Increased surface tension leads to rising surface energy rendering small bubbles less stable.

The three physical properties of the gas and the liquid phase are influenced by temperature and pressure and depend on the liquid composition (Table 4). An increase in temperature reduces liquid viscosity and surface tension significantly while its influence on gas density is less pronounced, therefore it tends to decrease stable bubble sizes, when operating pressure is high compared to the saturation pressure. Increasing operating pressure reduces surface tension slightly and increases gas density significantly. Both effects reduce stable bubble sizes.

Increased superficial gas velocity increases the bubble collision frequency leading to a higher coalescence rate and an increased stable bubble diameter. Liquid impurities such as alcohols and ketones can restrain coalescence not only for aqueous but also for organic gas liquid systems due to the formation of a boundary layer at the gas liquid interface.

Sparger design is crucial for such systems. Sparging behavior of different spargers was investigated with respect to initial bubble diameter (Table 5). Primary bubble sizes of porous spargers are mainly affected by wettability of its material, decreased pore diameters reduce initial bubble size only in non-coalescing systems. Bubble sizes of ring and capillary spargers decrease mainly with rising pressure and decreasing pore diameter, while coalescence inclination of the gas liquid systems shows no significant influence.

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